

Question Number	Answer	Additional guidance	Mark
2(a)(i)	The only correct answer is D $1.6 \times 10^3 \mu\text{m}$ <i>B is incorrect because the diameter is $1.6 \times 10^3 \mu\text{m}$</i> <i>C is incorrect because the diameter is $1.6 \times 10^3 \mu\text{m}$</i> <i>D is incorrect because the diameter is $1.6 \times 10^3 \mu\text{m}$</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(a)(ii)	A calculation showing the following steps: • correct volume from frog egg cell (1) • correct calculation (1) • correct calculated answer to 2 significant figures (1)	Mark the answer on answer line first <u>Example of calculation</u> $v \text{ of frog } (=4/3 \pi 800^3) = 2.14 \times 10^9$ $(2.14 \times 10^9) \div 1.8 \times 10^6 = 1189$ Allow ECF = 1200 times larger Correct answer (1200 or 1.2×10^3) to 2 sig figs with no working gains full marks	(3)

Question Number	Answer	Additional guidance	Mark
2(b)	An explanation that makes reference to four of the following: • egg cell contains {23 chromosomes/ half the genetic material of the skin cell} / egg cell is haploid (whereas skin cell is diploid) (1) • (therefore) fertilisation can occur to form a {diploid cell / zygote} (1) • (the egg cell genetic material is different from the skin cell) due to meiosis (1) • (the egg cell genetic material is different from the skin cell) due to crossing over (1) • (the egg cell genetic material is different from the skin cell) due to {random / independent} assortment (of homologous chromosomes) (1)	Accept {full set / original number} of chromosomes formed after fertilisation Accept fusion of nuclei Accept egg cell is formed by meiosis	(4)